

Fitness Tracker Product Discount Price Predictor

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Abstract

Modern lifestyle requires serious attention on fitness. Busy lifestyle more impact on the fitness need to be observed every time. Fitness tracker products resolved this issue which keep track on your fitness for you. Reason for this project is to identify the product with best discount price from variety of brands, color and other features. Models are designed to be predict the discount price therefor giving customers insight about the market which give advantage for buying

Predicting discount price requires series of steps and variety of methods for dealing with the process. For the project, two of the common models used namely linear and decision tree. Models requires several steps including data identifying and visualization, data pre-processing, hyper-parameter tuning and finally performance measuring and comparing for what is the best model.

I. INTRODUCTION

Fitness tracker which is a wearable device, is one of convenient products in human's life style as it helps to keep track of physical activities and monitor health including fitness-related metrics such as heart rate, steps taken, calories burned, and sleep patterns. Fitness trackers are available in the market with different brands, colors etc. But the price of them changes with different market affiliations. So that predict the discount price may be contained numerous cons in both business's and customer's side. For the businesses, it will helpful to stay in sync with the market and improve the effectiveness of their sales process. For some instances, it helps to understand the factors that influence the pricing of products and services, also it identifies the best time to offer discounts by targeting right customer at the right time. Not only that but also customers can gain maximum discount at the right time with the preferred product. Hence, to get the maximum advantage it will be helpful if there is a method to predict the discount price for a certain fitness tracker before buying.

In this project we are trying to create a best discount price predictor for fitness trackers in the market. To achieve that, we have used linear regression and decision tree supervised algorithms. Linear Regression algorithm is simple, interpretable and also capable of capturing linear relationships between features (Original price, Current price, Weight, Dial shape...etc) and the target variable (discount price). On the other hand decision trees are capable well of handling both numerical data (Original Price, Current price, Ratings, Display size...etc) and categorical data (Brand, Model Name, Dial Shape...etc). Also here, ensemble methods improve predictive accuracy by combining multiple decision trees.

II. METHODOLOGY

A. Data

Fitness Trackers Products Ecommerce from kaggle.com is the dataset used in the process. The dataset consists of information on fitness tracker products from various brands with their specifications and ratings on the Indian market. The dataset included variables such as product brand, current price, original price, discount percentage, ratings and also specification variables such as model name, dial shape, strap color, strap material, touchscreen availability, battery life, display size and weight.

In order to predict fitness tracker product discount price, variables was selected as importance to the prediction. That include numerical values such as current and original price for obtain market insights, ratings, battery life for customer influence and capability of product and also categorical values such as brand, model name, dial shape, strap color strap material for popularity and finer design which affect the discount price of the product.

TABLE I. DATA TYPES AND NULL VALUES OF FEATURES

Feature	Null values	Data type	Feature	Null values	Data type
Unnamed	0	Int64	Strap Color	123	Object
Brand	0	Object	Strap Material	69	Object
Current Price	10	Float64	Touchscreen	3	Object
Original Price	73	Float64	Battery Life	32	Float64
Discount Percentage	73	Float64	Bluetooth	6	Object
Rating	6	Float64	Display Size	30	Float64
Number of ratings	59	Float64	Weight	186	Float64
Model name	35	Object	Dial Shape	123	Object

B. Pre-processing

Pre-processing is done to improve the quality and readability of the data undergone for prediction model. The data set were subjected to various pre-processing methods which include treat missing values, removing outliers, converting strings in the dataset and also categorical features conversion, ANOVA.

Some variables such as display size, weight were which are strings were converted to the numerical by removing string and obtain float values. Additionally, weight range values substitute to obtain mean values and replace the existing range values. Also discount price variable which is

output variable, was derived from original price and discount percentage.

Numerical data, missing values found on number of variables such as current price, original price, rating, number of ratings, battery life, display size, weight and discount price, were handled using imputation using median values. Selection of median value for the imputation is to ensure data distribution throughout the dataset. In the categorical data, missing values were addressed by imputing placeholder value ("Other").

Outliers in the dataset was identified using visualization methods (Box plot) and addressed them using IQR method which removed them from data. Then numerical values were subjected to feature scaling which normalize the numerical values for have the same scale. This method increases the sensitivity of the model increasing accuracy. For the categorical values, first ANOVA is done to understand the variance of the output variable "Discount Price" in different categories. After that values were subjected to one-hot encoding which used to convert distinct values into new columns giving them binary values.

The dataset was split into two parts, one for training and other set for testing. Parts were separate 80% and 20% from total dataset for training and testing respectively. Train-test-split techniques used for dividing the dataset which randomly divide the dataset according to the random state value given.

C. Algorithms

In this project the discount price of the fitness tracker was the target variable. Since the target variable is continuous one and it was assumed that the Discount Price has a linear relationship with other features of the fitness tracker, Linear Regression Algorithm and Decision Tree Regression Algorithm. Both Linear Regression Algorithm and Decision Tree Regression Algorithms are Supervised Learning as they are used to predict continuous output variables based on input features in the data set.

When creating a fitness tracker discount price predictor, as the discount price is the target variable, the linear regression algorithm is essential to the process. A linear relationship is established between independent variables, or features, and the dependent variable, or discount price that according to this project, using the statistical modeling approach known as linear regression. Numerous factors, such as sophisticated features, brand recognition, and consumer demand, might affect the discounted pricing in the context of a fitness tracker. Based on the input features, the linear regression model predicts future discount prices by analyzing previous data to find patterns and relationships. We were able to use this algorithm to obtain insights into the aspects that influence fitness tracker pricing strategies, which helps us make well-informed decisions when determining how much to discount in order to increase customer engagement and market competitiveness.

The linear Regression equation can be expressed as,

$$Y = b_0 + b_1 X_1 + b_2 X_2 + \dots + b_n X_n + \epsilon$$

Y- Discount Price

b₁, b₂,.....b_n- Coefficients for the predictor variables(X₁,X₂,....., X_n)

The aim of the linear regression is to find the coefficients that minimize the sum of squared difference between actual and predicted values.

Cost Function J(θ)=squared error function

$$J(\theta) = \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^i) - y^i)^2 \quad [1]$$

When it comes to Decision Tree Regression, it is also a machine learning approach that may be applied to situations where the objective variable is numerical, such estimating the discounted pricing in this fitness tracker project. The process of decision tree regression involves building a tree-like model that divides the dataset recursively according to feature values. Each terminal node is then given a predicted value, according to in this project, the discount price is given at the end. As described above, relevant features for fitness trackers could be things like company reputation, market trends, and technological requirements. By evaluating the effects of various characteristics at each node and optimizing splits to reduce variance in predicted values, the decision tree algorithm gains the ability to anticipate future events. The resulting tree structure does a good job of capturing the intricate connections between discount pricing and features. The interpretability of this model makes it possible to clearly comprehend the variables affecting fitness tracker pricing selections.

The fundamental equation for decision tree can be expressed as,

$$y = f(X_1, X_2, \dots, X_n)$$

y : Predicted discount price.

X₁, X₂, ..., X_n: Features related to the fitness tracker

D. Implementation

The code of the linear regression algorithm was constructed based on a videos and other electronic resources.

Scikit-learn library in python, which is a common library for efficient tools in various algorithms was utilized for the model implementation.

Below it shows the importing necessary libraries and initialization of each algorithm accordingly,

Linear Regression Model

```
from sklearn.linear_model import
LinearRegression,Ridge, Lasso,
lr_model = LinearRegression()
```

Decision Tree Regression Model

```
from sklearn.tree import DecisionTreeRegressor
tree_model = DecisionTreeRegressor()
```

The basic code for both linear and decision tree regression algorithms were modified minimally. To address missing data, encode categorical variables, and scale features, pre-processing techniques were used. Feature engineering had been used to improve the models' capacity for prediction.

Grid search method was applied to discover the best configuration and adjust hyper-parameters for both linear regression and decision tree regression. Variations were made to parameters like regularization strength for linear regression and also max depth, min samples split and min samples leaf for decision tree regression. The obtained values for those parameters can represent as follow,

Regularization strength = 0.001
 Max depth = none
 Min samples split = 2
 Min samples leaf = 4

Then based on the results of the grid search for final model training, the hyper-parameters were fixed.

III. RESULTS

Both models are used for predicting the test data individually after that results are obtained using R squared value which is coefficient determination. Used to measure variance of the dependent variable which can be explained by the independent variable. The R squared value used measure the performance of the model, closer to 1 better the model fit the data.

TABLE II. R SQUARED VALUES FOR EACH MODEL

Measure	Linear Regression	Decision Tree Regression
R Squared error	0.8704159597695442	0.6597746722349793

As shown in the “TABLE II”, linear regression model produced rounded R squared score of 0.87 while the decision tree regression model produced rounded R squared score of 0.66. And also observed when computing prediction value time taken for both models are significantly difference. Time for calculating predictions for linear regression model is lower than time taken for decision tree regression model in calculating predictions.

Additionally, for decision tree regression model, we have obtained additional values for measuring performance and the accuracy of the model like Cross-validated score and mean cross validated score. Cross-validation is a method of estimating performance skill of the model which divided the dataset into number folds (k) and then train model using k-1 folds used as training and remaining fold as the test data. This method reduce bias in the system and reducing variance of the data. For this process, we have taken 5-fold cross validation and resultant scores of cross validated as follows,

[0.46251607, 0.6459738, 0.00335857, -0.16506898, 0.64184775]

And the mean cross validated score of 0.31772544.

IV. DISCUSSION

A. Conclusion

For the conclusion regarding the performance of the linear and decision tree regression algorithms, we have consider the R squared value obtained for each model. As shown in “TABLE II”, R squared value for linear regression model is closer to 1 than decision tree regression. Therefore, for the

current dataset linear regression model fit well than the decision tree model.

As for the decision tree model, cross validation scores obtained for 5 folds shows that 1 fold is showing negative cross value which means that model perform poorly on those particular data. Reason for this behavior could be because of data on that fold 4 could be different from other folds. And also because of the model overfitting or model to be too complex. But also we can see noticeable variance in cross values which also signed that model cold be over fitted.

But other folds show significant good results. Therefor we can conclude that the decision tree regression model is moderately accurate on the dataset.

By considering overall performance of the both models, we can conclude that by comparing linear regression model perform better that decision tree regression model and have better accuracy in predicting.

B. Ethics

Predicting the discount price benefits both customer and the producers. Use of these machine learning models on real life situation requires several ethical key aspects.

Security: Security is the most important consideration in machine learning ethics, requiring strict precautions to protect models and data. Important elements of ethical machine learning security include strong access limits, secure storage options, and encryption techniques. The implementation of these protections is done with great care to prevent malicious exploits and illegal access, protecting sensitive data integrity and confidentiality. Ethical machine learning methods prioritize security, which builds user trust and strengthens AI systems resilience. This builds a basis for responsible and safe machine learning technology implementation across a range of disciplines.

Privacy: In ethical machine learning, privacy plays a crucial role by carefully balancing the needs of data utility with the privacy of the individual. Strict access limits, transparent data processing, and careful data anonymization are examples of ethical privacy practices. By taking these precautions, users may feel confident that machine learning models are being deployed ethically and that their personal information is handled responsibly, all while upholding their right to privacy.

Fairness and Bias: Fairness and bias mitigation are essential components of ethical machine learning. In order to provide fair results, ethical standards entail locating and eliminating biases in training data and algorithms. In order to generate fair and impartial outcomes, ethical machine learning aims to eradicate bias from model predictions. Fairness and bias concerns aid in the creation of AI systems that respect moral principles and encourage objectivity in a range of applications by means of strict examination and remedial actions.

Transparency and explain ability: Essential components of ethical machine learning include explain ability and transparency, which provide insights into model choices. In order to promote accountability and user trust, ethical practices place a high value on transparent and intelligible explanations. Transparency makes machine learning models easier to interpret, enabling people to understand and question results. By guaranteeing responsible deployment and fostering

a more knowledgeable and trustworthy relationship between users and machine learning technology, this dedication to openness promotes ethical AI development.

Accountability and Responsibility: Accountability and responsibility are the cornerstones of conscientious development and application in ethical machine learning. Ethical standards and well defined responsibilities guarantee that system conduct is accountable to all parties involved. By reducing the possibility of misuse and unforeseen effects, this commitment promotes the responsible application of AI. In addition to providing protection against moral lapses, ethical accountability creates a framework that encourages the moral application of machine learning technology with an emphasis on openness and user confidence.

Continuous Learning and Adaptability: In ethical machine learning, it is essential to be flexible and always learning. Models can adapt responsibly to changes in data distribution and societal expectations by receiving regular updates and close observation. This dedication promotes a flexible and adaptable AI environment by guaranteeing continued ethical

performance and bringing machine learning algorithms into compliance with changing ethical norms.

V. REFERENCES

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